

Computer Vision

Machine Learning

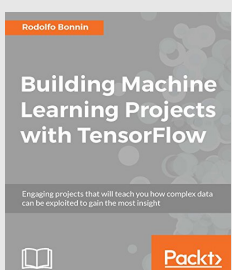
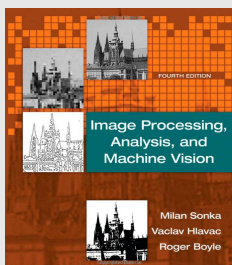
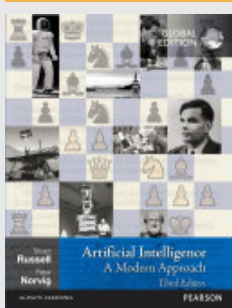
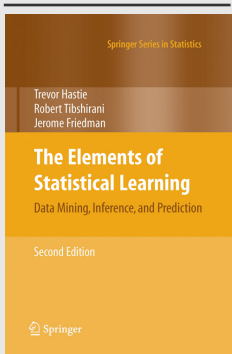
Big Data

Artificial Intelligence

Parallel Computing

Numerical methods

BooksTaught



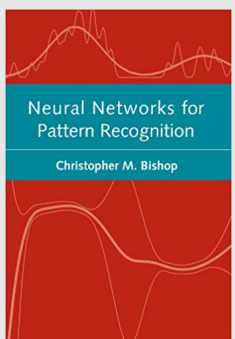
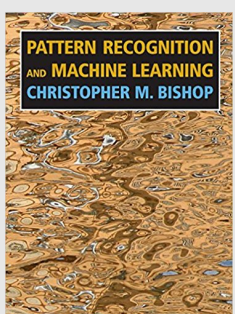
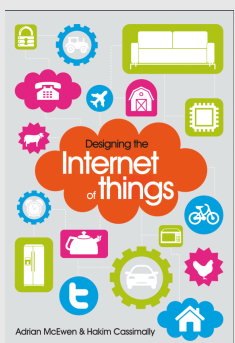
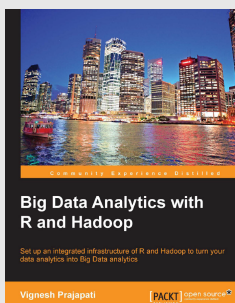
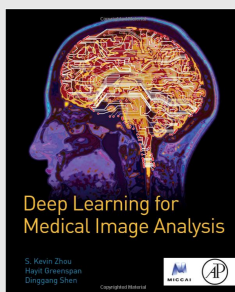
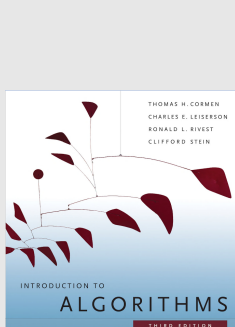
Teaching Statement

Academia provides an opportunity to combine research and teaching in a way that is not possible in any other setting: to teach classes, get students engaged in research, and ultimately retro-feed some aspects of that research back into the classroom. I attempt to take my teaching responsibilities to heart, both by considering how to take advantage of the interaction between teaching and research, as well as by continually re-evaluating how trends in industry should reflect upon my teaching. I enjoy teaching mainly for two reasons: one reason is self-enrichment and the other being a complement to my research. I consider myself both a “partner” and a “perpetual learner” in the pedagogic process. Teaching has also offered me the most rewarding professional experiences as I have seen students’ eyes light up while comprehending a new concept.

Teaching Philosophy: As an educational architect, I foster the essence of learning with great enthusiasm by creating a constructive, interactive, cooperative and collaborative environment that facilitates learning. My teaching philosophy centers around three perspectives. First, I make sure that the classrooms incorporate a diversity of learning techniques that encourages a diversity of students. While I teach, I don’t think that I teach “a class of students”, rather I teach “a set of unique individuals”. I meet with each of my students on a one-on-one basis during the first month of each semester to facilitate a diversified approach to learning course topics and of planting them on a wide spectrum so that it could satisfy the interests of students from diverse backgrounds. Second, while I teach I don’t confine myself only to textbook materials, rather I demonstrate the “big picture” of the topic, experiential case studies, ongoing research, industry applications and probable future research directions. Experiential learning provides greater insights and longer retention than textbook materials. Third, I do not believe in hand-holding the students. It is my experience that setting firm and high expectations, and carrying through on these expectations motivates students to rise to the occasion, resulting in students working more effectively and therefore learning more. I choose content and activities that are both significant and challenging for the students so that they are stimulated to think “beyond” their comfort zone and feel involved in their own learning process. Similarly, I facilitate the activities and class discussions that encourage students to make connections between my field of study and other disciplines. Collaborative learning is core to my teaching philosophy in order to promote reflective thinking and improve the students’ communicative and trans-cultural skills. I try to create “classrooms without walls” where each student gets the opportunity of developing his ability to deal with the real world challenges. I encourage students to take responsibility and hold them accountable for their actions and performance so that they can lead their work in future.

Teaching Methodology: I always strive to provide the intuition behind the techniques and theories being presented, and to draw the students’ attention to the fundamental concepts underlying the material taught. In addition, when presenting basic mathematical concepts or abstractions some students may find intimidating, I try to emphasize how a deeper understanding may ultimately help the students tune or debug an application and also I seek to identify the potential applications of these concepts in the solution of real-world problems. I always found that motivating a topic from a historical perspective and the basic understanding helps immensely in stimulating students’ zest and encourages questions and discussions. I put emphasis on problem formulation as much as on than problem solving itself. One of my goals as a teacher is to get students to think critically, instilling on them a taste for challenging what they are taught. I favor an accessible, “semi-structured” and interactive teaching style that invites student participation. Guided class discussions encourage the students to think of potential solutions to a given problem. Like many of my colleagues, I use PowerPoint slides to help organize and summarize lecture material. However, my slides have portions that are deliberately left blank, sometimes filled with erroneous input and must be filled or corrected in during class discussions, or worked out on the board with student input. I also like to point students to webpages or blogs where they can see more material, because this can reinforce the value of the topic. Furthermore, I try to embrace technologies which facilitate discussion, interaction, and collaboration, whether they be traditional means such as message boards or more recent tools including online chat rooms or clickers in the classroom. The application of technology and computer-based material in the classroom such as Webboard, Wikis and YouTube are helpful tools in the promotion of meaningful learning through collaborative work as well.

Professional Development: I have successfully completed a formal teaching training curriculum conducted at University of Alberta (UofA), Instructional Skills Workshop (ISW) at University of Calgary and Project Management Workshop (PMW) organized by MITACS which were useful for honing my teaching style. University Teaching Services (UTS) consist of 20 formal Pedagogy lectures combined with supervised in-class practicum in regular courses. The program included a series of seminars which have proven to be valuable resources, as well as teaching observations followed by feedback session, which develops useful teaching skills. ISW offered hands-on experience on designing mini-lessons using BOPPPS teaching model, performing learning style survey before preparing lessons for large-room classes, gaining knowledge of effective questioning that enhanced student learning, and creating a positive learning environment in the classroom along with the successful use of the current web-based e-learning technology called “Top-Hat”. PMW offered hands-on experience on Strength Deployment Inventory (SDI) and card on the wall method, presentation skills which engage audience through authenticity, and body language, as well as the art, science and practice of positive networking. I regularly watch lectures on teaching methodologies in TEDTalk and youtube.



Teaching Experience: At present, I am teaching Introduction to Software Development (CMPT 211) and Introduction to Information Technology (IT 101) at Concordia University of Edmonton (CUE). In the last term I taught Introduction to Computing Science (CMPT 111) at CUE. Previously, I taught Fundamentals of Big Data: Tools and Algorithms, Medical Image Processing, Numerical Methods and Optimization for Big data, Computational Statistics, Numerical Methods and Optimization, Bioinformatics, as well as Programming and Analysis of Algorithm at graduate level in Centro de Investigación en Matemáticas (CIMAT), Mexico. In addition, I taught Artificial Intelligence and its application in Robotics and Manufacturing, Machine Learning and Big Data at graduate level in Centro de Ingeniería y Desarrollo Industrial (CIDESI), Mexico. During my graduate studies at University of Alberta, I worked as a Teaching Assistant (TA) for three years. There I taught a basic undergraduate course Introduction to Computing (CMPUT 101), a mid level undergraduate course 3D Graphics and Animation with 3DS Max (CMPUT 307), a senior undergraduate/graduate level course Introduction to Numerical Methods (CMPUT 340/418/670) and a graduate course Application of Machine Learning in Image Processing (CMPUT 615). I have given several talks to diverse audience in various institutes as well as in different international conferences across the world. The talks given at reading groups, tutorials, research conferences or in a classroom are nothing but different forms of teaching. The difference is in the background of the audience, and the duration of presentation.

Evaluation of Teaching: The feedback I received from my classes was very positive. For example, when asked to evaluate my overall effectiveness as a teacher, 97% of my students rated my performance as excellent or good. I received an average 9.5 out of 10 rating on the student evaluations. I designed a very challenging course and was extremely satisfactory with how well the class did. However, I received positive rewarding (and ongoing) experiences from class projects which led to four journal and three conference publications in the last two years.

Advising: At present, I am mentoring two PhD, three master and two undergraduate students having multidisciplinary background from multiple institutions. I have already supervised two postdoctoral research fellow, three master, and eight undergraduate students. I published several research papers with students as a co-author. I interact with them frequently (at least once in a week individually) and I held a group meeting once in every week. As an adviser, I try to formulate the long-term goals considering the big picture and the short-term goals considering the realistic targets. I was given the opportunity to supervise students from National Institute of technology, India. I remotely advised students helping in framing and understanding of the problems related to their thesis. During summer, I mentor undergraduate students to develop prototypes that involve artificial intelligence. Last summer, under my supervision two of my students from the Mechatronics department developed Arduino based tank water level and temperature indicators as well as automatic pump controller using kalman filter and camera calibration, as well as a Raspberry based robot controller using kalman filter multi-sensor combination algorithm. I advised students in an industrial workshop regarding how to implement statistical sampling plan for visual inspection of electronic chip making processes.

Curriculum Innovations: I have served on curriculum innovation committees for designing a Masters program in Information Technology at CUE, and Computational Statistics in CIMAT. I am lucky to have taken part in a team of experts for designing a new curriculum on Information Technology at CUE and Computational Statistics, which was the first of its kind in Mexico. Many Universities didn't start to teach Computational Statistics even when the subject was under high demand in the academia and industry.

Education Outreach: In my opinion, teaching never ends in the classroom. Hence out-of-the-classroom education is perfect complement to classroom learning and a way to help the community. I am fortunate to have had the experience of working with physicians in different areas of medicine and with biologists.

Teaching Interests: My diverse educational background (Bachelor of Mechanical Engineering, Masters in Quality, Reliability and Operations Research (QROR) as well as a Masters and PhD in Computer Science) along with post-doctoral research in Canada and United States strongly equips me to teach a wide range of courses on Computer Science, Statistics, Quality, Reliability, Optimization, Mechanical Engineering, and interdisciplinary areas. I would also like to develop new courses in accord with departmental needs.

Future Goals: Based on my past experience, I found all aspects of teaching to be stimulating and rewarding: lecturing, interacting with students, formulating problem sets, lab experiments and advising. My aspiration is to become a Professor at a quality research-oriented university. I believe that excellence in teaching is not a destination, it is a never-ending journey. I observe my peers and learn how they approach in matters of engaging students in and outside the classroom. I encourage the suggestions of other faculty members and comments from students so as to improve my style of teaching. I regularly take part in the workshops provided by the University Teaching Center and attend online courses that continually offer very useful ideas and enhance my teaching proficiency. I view teaching as a series of experiments and each such experiment I consider as a "learning opportunity". I believe in adopting new techniques and technologies in the classroom such as would enhance learning environments, and I am excited to see where the evolution of university teaching would take take us in the future.